



Deciphering Seasonal Weather Impacts on Crash Severity: A Machine Learning Approach

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HSIS First Place Safety Data Award

This is the first-place winning paper of the Federal Highway Administration's (FHWA) 2024 Excellence in Highway Safety Data Award, which is designed to encourage university students to use Highway Safety Information System (HSIS) data to investigate a topic that advances highway safety and to develop a paper to document the original research. The HSIS Highway Safety Data Awards Program is jointly administered by FHWA and ITE. This paper has been edited for length from the original version.

Traffic safety remains a critical concern, especially under adverse weather conditions that significantly affect driving dynamics and road safety. The U.S. Department of Transportation (USDOT) defines weather-related crashes as those occurring in conditions such as rain, snow, fog, and severe crosswinds. In Minnesota, the climate varies dramatically, with seasons ranging from frigid Winters to humid Summers. This diverse climate necessitates an understanding of how weather conditions affect crash severity to enhance traffic safety and reduce weather-related road incidents.¹

Studies in Minnesota highlight the high stakes of weather impacts, noting that severe weather events like tornadoes can lead to significant property damage and increased costs.² Precipitation in Winter is associated with a 19 percent increase in traffic crashes and a 13 percent increase in injury crashes compared to dry conditions.³ The Minnesota Department of Transportation (MnDOT) has explored various mitigation strategies, such as optimizing gate operations for clearing highways, which have been shown to reduce costs by 15 percent.⁴ Advanced Intelligent Transportation Systems (ITS), such as weather sensor-equipped connected vehicles, provide site-specific road weather data guiding motorists in real-time.^{5,6}

This study investigates how seasonal weather conditions affect crash severity in Minnesota. By examining the individual and combined effects of weather and seasonal variations, the study provides a detailed analysis of how adverse weather conditions influence crash outcomes. This study is especially pertinent as adverse weather not only leads to delays and higher crash rates but also significantly affects road traffic injuries by limiting visibility, increasing stopping distances, and creating hazardous road surfaces due to ice or wetness.^{5,7,8}

Methodology

Data Collection

Data were sourced from the Highway Safety Information System (HSIS). HSIS has extensive coverage and detailed data quality. It encompasses diverse variables that capture environmental, temporal, and situational factors influencing traffic incidents. Utilizing HSIS data allows for an in-depth exploration of crash incidences, considering Minnesota's variable climate and unique geographical challenges. The database's rigorous standards for data collection and validation ensure the reliability of our findings, facilitating an accurate assessment of correlations between weather, seasonal shifts, and crash severity. This study analyzed crash records from Minnesota from 2017 to 2019, comprising 25,880 observations, capturing various elements such as crash details, weather conditions, light conditions, and crash location.

Data Preparation

Data were preprocessed to ensure quality and consistency. Missing values were addressed based on their nature and occurrence in the dataset. Variables were converted into appropriate formats for analysis.

Minnesota categorized crash severity using the KABCO format. For analysis purposes, these severity levels were grouped into three categories. "Minor" for no apparent injury, "Moderate" for possible (C) and non-incapacitating (B) injury, and "Severe" for serious (A) and fatal (K) injury. Crash data was grouped into four seasons: Spring (March, April, May), Summer (June, July, August), Autumn/Fall (September, October, November) and Winter (December, January, February). Table 1 shows the descriptive statistics.

Table 1: Descriptive Statistics

Variables	Categorization	Frequency	Percent
Weather	Clear	21,655	83.67%
	Fog/Smog/Smoke	68	0.26%
	Rain	1,862	7.19%
	Snow	2,295	8.87%
Season	Autumn	6,177	23.87%
	Spring	5,981	23.11%
	Summer	6,772	26.17%
	Winter	6,950	26.85%
Intersection Related	No	4,653	17.98%
	Yes	21,227	82.02%
Light Condition	Dark	5,449	21.05%
	Daylight	20,431	78.95%
Crash Severity	Minor	18,711	72.30%
	Moderate	3,936	15.21%
	Severe	3,233	12.49%

Model Development

Assessment of Association Between Response Variable and Predictors

A chi-square test of independence was performed to investigate the association between the predictors and crash severity. The chi-square test determines if the likelihood of items being classified for one variable depends on the other. A growing concern has been shifting toward the necessity of improving traffic safety in the Kingdom of Saudi Arabia (KSA).^{9,10}

The chi-square formula:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (1)$$

where O_i is a set of observed frequencies and E_i is the corresponding set of expected frequencies.

Multinomial Logistic Regression

To assess the impact of seasonal weather conditions on crash severity, the study utilized multinomial logistic regression (MNL), aligning with several previous studies including the roadway, driver, traffic characteristics, and the environment. This study is concerned with the outcome of the crash. Method: Driver injury severity levels are analyzed using the ordered probit modeling methodology. Models were developed for roadway sections, signalized intersections, and toll plazas in Central Florida. All models showed the significance of driver's age, gender, seat belt use, point of impact, speed, and vehicle type on the injury severity level. Other variables were found significant only in specific cases. Results: A driver's

violation was significant in the case of signalized intersections. Alcohol, lighting conditions, and the existence of a horizontal curve affected the likelihood of injuries in the roadway sections' model. A variable specific to toll plazas, vehicles equipped with Electronic Toll Collection (ETC).¹¹⁻¹³ The model classified crash severity into three categories: severe, moderate, and minor, which served as the response variable. Predictor variables included weather, seasons, intersection involvement, and light conditions. Coefficients were estimated through maximum likelihood, and goodness-of-fit was evaluated using likelihood ratio tests. The MNL model was fitted function in *R*.

The MNL model:

$$P_{ni} = \frac{e^{\beta_i X_{ni}}}{\sum_{i=1}^I e^{\beta_i X_{ni}}}, i = 1, 2, 3 \quad (2)$$

where P_{ni} is the probability of the crash n to suffer the output injury of i ,

β_i is a vector of the calculative coefficient for the output severity and

X_{ni} is a vector for the explanatory variables.

Random Forest Analysis

In addition to the MNL model, the study applied the Random Forest (RF) method as a classification technique to derive a ranking of variable importance based on classification accuracy and the Gini impurity coefficient. RF is known for its robust performance across various prediction problems and its ability to handle non-linear effects and interaction terms without prior specification.¹⁴⁻¹⁶ The combined approach leverages the strengths of each method to provide a comprehensive view of the factors influencing crash severity under various seasonal weather conditions.¹⁷

Validation and Robustness Testing

A cross-validation approach was employed to evaluate the predictive performance of the MNL model. The dataset was divided into ten equal-sized folds, and the model was trained on nine folds while tested on the remaining fold iteratively. This method assessed the model's accuracy and robustness across different data subsets. The independence of irrelevant alternatives (IIA) was tested using the Hausman-McFadden test, which validated the model's appropriateness for this study.¹⁸

Results and Discussion

Visualization

Figure 1 shows the correlation matrix indicating correlation coefficients between various predictor variables. Variables with a correlation coefficient above 0.5 are collinear.¹⁹ None were found to correlate to one another.

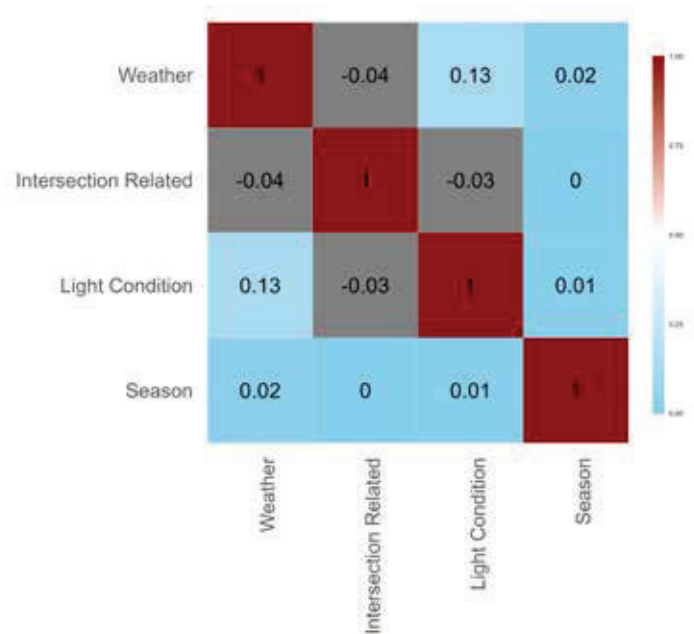


Figure 1: Correlation Matrix

Figure 2 shows a heatmap displaying the normalized distribution of crashes by weather and season. The values in each cell indicate the proportion of crashes occurring under each combination of weather and season, normalized by the total number of crashes for each weather condition. Clear weather shows an even distribution of crashes across all seasons, with a slightly higher frequency during Summer (0.29). Fog has a dominant occurrence of crashes in Winter (0.68), pointing to significantly adverse conditions during the season. Rain sees the highest proportion of crashes in Autumn (0.43), followed by Summer (0.25). Snow, predictably, results in most crashes during Winter (0.58), with no crashes reported during Summer simply because it can never snow in Summer.

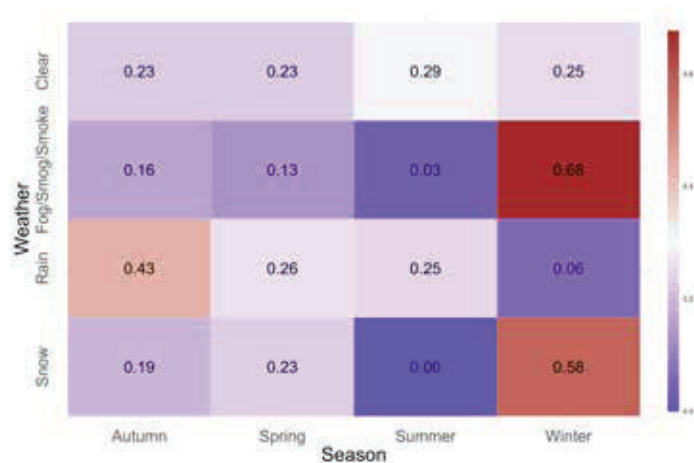


Figure 2: Heatmap of Crash Severity vs. Weather and Season

Model Results

The evaluation of MNL and RF models shows that the fitted model achieved an accuracy of 72 percent in the prediction of the weather-related factors influencing the crash injury severity levels. The Area Under the Receiver Operating Characteristic (AUROC) values for severe, moderate, and minor injuries were 0.565, 0.597, and 0.590, respectively, as presented in Figure 3.

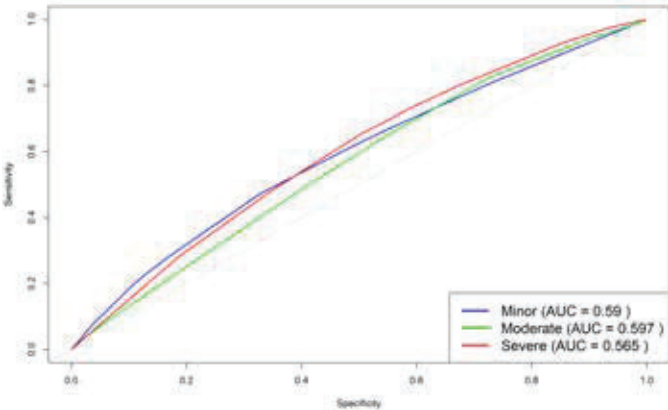


Figure 3: Multiclass ROC Curves for Crash Severity Model

Table 2 presents the results of the bivariate analysis conducted to determine the association between various predictor variables and crash severity at a 95 percent confidence interval (CI). The null hypothesis states that there is no association between the specific predictor and the crash severity. On the other hand, the alternative hypothesis states that there is an association between the predictor

and the response variable. The *p*-value for light condition exceeds 0.05, indicating no significant association with crash severity. Additionally, a variable importance analysis using a RF model was performed. The analysis revealed that the lighting condition has the least importance in explaining the injury severity, leading to the variable exclusion from the final model.

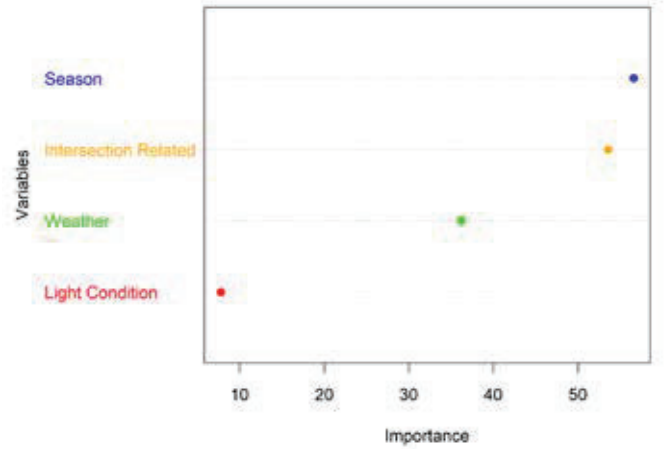


Figure 4: Variable Importance Plot from RF Model

Table 3 presents the MNL model results. *P*-values indicate the significance of the estimates, with values less than 0.05 generally considered statistically significant. Relative risk ratio (RRR) values greater than 1 indicate an increase in risk relative to the reference category. In contrast, values less than 1 indicate a decrease in risk relative to the reference category.

Table 2: Bivariate Analysis of Crash Severity and Different Factors

Factors	Categorization	Crash Severity								P Value
		Minor		Moderate		Severe		Total		
		Freq	%	Freq	%	Freq	%	Freq	%	
Weather	Clear	15,480	82.73	3,334	84.71	2,841	87.88	21,655	83.77	0.00***
	Fog/Smog/Smoke	45	0.24	11	0.28	12	0.37	68	0.26	
	Rain	1,303	6.96	336	8.54	223	6.90	1,862	7.20	
	Snow	1,883	10.06	225	6.48	157	4.86	2,265	8.76	
Season	Autumn	4,347	23.23	1,014	25.76	816	25.24	6,177	23.87	0.00***
	Spring	4,368	23.34	874	22.21	739	22.86	5,981	23.11	
	Summer	4,557	24.35	1,110	28.20	1,105	34.18	6,772	26.17	
	Winter	5,439	29.07	938	23.83	573	17.72	6,950	26.85	
Intersection Related	No	3,758	20.08	468	11.89	427	13.21	4,653	17.98	0.00***
	Yes	14,953	79.92	3,468	88.11	2,806	86.79	21,227	82.02	
Light Condition	Dark	3,980	21.27	790	20.07	679	21.00	5,449	21.05	0.24
	Daylight	14,731	78.73	3,146	79.93	2,554	79.00	20,431	78.95	

Note: **p*<0.1; ***p*<0.05; ****p*<0.01

The findings indicate that among the three weather categories, rain and snow were found significant at a 95 percent CI with a negative association with injury severity. In addition, all three seasons and intersection-related crashes were found to be significant at a 95 percent CI. Regarding the combined effects of weather and seasons, findings indicate that fog-spring, fog-summer, and snow-winter were significant at a 95 percent CI.

Weather Conditions

Fog, smog, or smoke significantly increase the likelihood of crashes with severe injuries (RRR=2.00), implying more severe outcomes when visibility is reduced. However, these conditions are less associated with moderate injuries (RRR=0.57), indicating that while visibility issues may escalate the severity of crashes that do occur, they may not uniformly increase the frequency of less severe crashes.

Rain reduces the risk of severe crashes (RRR = 0.75). Thus, rainy weather has a 25 percent lower probability of resulting in

severe crashes compared to clear weather conditions. This could be due to increased driver alertness during rainy conditions (20–22) the act of walking is sensitive to changes in weather conditions. However, comprehensive real-time weather data are often unavailable for road safety analysis. In this study, we used a geographical information system approach to integrate high-resolution weather data, as well as their corresponding temporal and spatial distributions, with crash data. Then, we established a mixed logit model to determine the association between pedestrian crash severity and possible risk factors. The results indicate that high temperature and the presence of rain were associated with a higher likelihood of Killed and Severe Injury (KSI).

Snow reduces the likelihood of both moderate (RRR=0.57) and severe crashes (RRR=0.33). This might be explained by reduced vehicle miles traveled during heavy snowfall as drivers avoid hazardous conditions.

Table 3: MNL Model Results

Variable	Factors	Crash Severity					
		Moderate (BC)			Severe (KA)		
		Estimate	Std. Error	RRR	Estimate	Std. Error	RRR
	Constant	-1.98	0.06	0.14	-2.03	0.06	0.13
Weather	Clear						
	Fog/Smog/Smoke	-0.56	1.07	0.57	0.70	0.69	2.00
	Rain	0.11	0.10	1.12	-0.29**	0.12	0.75
	Snow	-0.56***	0.16	0.57	-1.13***	0.21	0.33
Season	Autumn						
	Spring	-0.13**	0.06	0.88	-0.12**	0.06	0.88
	Summer	0.02	0.05	1.02	0.18***	0.05	1.20
	Winter	-0.27***	0.06	0.76	-0.63***	0.06	0.53
Intersection-Related	No						
	Yes	0.635***	0.05	1.89	0.54***	0.06	1.71
Weather & Season	Clear & Autumn						
	Fog/Smog/Smoke & Spring	-9.28***	0.00	0.00	-0.28	1.06	0.76
	Rain & Spring	-0.06	0.17	0.94	0.13	0.19	1.14
	Snow & Spring	0.15	0.21	1.16	0.44	0.27	1.56
	Fog/Smog/Smoke & Summer	-9.56***	0.00	0.00	-13.39***	0.00	0.00
	Rain & Summer	0.10	0.16	1.10	0.21	0.18	1.24
	Snow & Summer	0.00	0.00	1.00	0.00	0.00	1.00
	Fog/Smog/Smoke & Winter	1.18	1.13	3.24	0.08	0.81	1.08
	Rain & Winter	-0.20	0.31	0.82	0.47	0.35	1.61
	Snow & Winter	0.22	0.18	1.25	0.85***	0.24	2.34

Note: Bolds are the base category. *p<0.1; **p<0.05; ***p<0.01. Minor is the base category for crash severity.

Seasons

Spring and Winter are associated with less severe crashes compared to Autumn. Crashes in Spring and Winter are less likely to result in severe injury crashes by 12 percent and 47 percent, respectively.

Summer is associated with more severe crashes, with a 20 percent greater likelihood of resulting in fatalities or serious injuries compared to Autumn. This could be due to higher traffic volumes, increased travel speeds, or more outdoor activities during warmer months.²³

Intersection-related Crashes

Intersection-related crashes have an increased likelihood of resulting in moderate and severe crashes by 89 percent and 71 percent, respectively, compared to non-intersection-related crashes. Intersections are attributed to more angle crashes.^{24,25} The severity of intersection-related crashes may be attributed to most of these crashes involving pedestrians and bicyclists due to their vulnerability.^{25,26} Other studies have also identified the severity of left-turn crashes at intersections, making intersection-related crashes more severe.^{26,27}

Combined Effects of Weather and Season

Foggy conditions in Spring showed a significantly negative association with injury severity, suggesting a mitigation effect possibly due to reduced vehicle speeds or heightened driver caution. Foggy weather in Summer also showed a pronounced negative association with injury severity, likely because foggy conditions are rare in Summer.

The combined effect of snow conditions in Winter significantly increases the likelihood of severe crashes ($RRR=2.34$). This highlights the compounded challenges posed by snow conditions in Winter, such as poor visibility and slippery roads.

Conclusions and Recommendations

This study utilized HSIS data to examine how weather and seasonal changes affected crash severity in Minnesota from 2017 to 2019. The findings highlight that both individual and combined weather and seasonal factors significantly influence crash severity. Fog in Spring and snow in Winter notably increase the likelihood of severe injury crashes (KA). Understanding these dynamics underscores the critical role of specific weather-season combinations.

Fog increases the likelihood of severe injury crashes, particularly in Spring, while rain generally reduces severe crashes due to heightened driver caution. Snow can lead to severe crashes, although lower traffic during heavy snowfall mitigates some risks. Spring and Winter are associated with lower severity crashes compared to Autumn, whereas Summer increases the likelihood of severe crashes due to higher traffic volumes and speeds. Intersection-related crashes also tend to be more severe.

Recommendations include implementing fog alert systems and enhancing driver education for foggy conditions during Spring, optimizing de-icing methods for Winter, and correlating crash data

with specific maintenance practices. Before-after studies should evaluate the effectiveness of any local weather-related ITS improvements. Integrating these insights into traffic management systems will enable targeted interventions, ultimately reducing crash severity and enhancing safety in Minnesota. [itej](#)

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Answer to "Where in the World?" on page 19: Philadelphia, PA, USA. Photo submitted by Kellyanne Broom.